

BIOS 755: Longitudinal versus Cross-sectional Effects and Covariance Patter versus LMM

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Introduction

- ▶ It is possible to allow for **longitudinal** and **cross-sectional** effects in longitudinal analyses.
- ▶ Such an approach acknowledges the two distinct sources of variation in a covariate:
 - ▶ one based on within-subject variation, and
 - ▶ one based on between-subject variation.
- ▶ Such a model recognizes that longitudinal data provide information about
 - ▶ how individuals differ at any one occasion, and
 - ▶ how an individual's response varies over time.
- ▶ Commonly, these effects are erroneously combined.

Example

- ▶ Suppose there is a study on the impact of talkativeness and well-being (subjectively measured).
- ▶ There could be a positive association between talkativeness and well-being between-people:
 - ▶ people who are (on average) more talkative than others are (on average) happier than others.
- ▶ There could be a positive association between talkativeness and well-being within-people:
 - ▶ people who are more talkative today (than they usually are) are happier today (than they usually are).

Example

- ▶ **These effects may be equal,**
 - ▶ if a person is 20% more talkative (either between or within), we predict they will score 15% higher on their well-being (on average).

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- ▶ **These effects may be different,**
 - ▶ if person A is 20% more talkative than person B (on average), we predict person A will score 5% higher than person B on their well-being (on average), while
 - ▶ if person A is 20% more talkative today (than they usually are), we predict they will score 20% higher than usual on their well-being (on average).

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 - ▶ if person A is 20% more talkative today (than they usually are), we predict they will score 20% higher than usual on their well-being (on average).
- ▶ **These effects may be the opposite,**
 - ▶ if person A is 20% more talkative than person B (on average), we predict person A will score 5% higher than person B on their well-being (on average), while
 - ▶ if person A is 20% more talkative today (than they usually are), we predict they will score 10% lower than usual on their well-being (on average).

Example

- ▶ **Between person effects**
 - ▶ **if person A is 20% more talkative than person B (on average), we predict person A will score 5% higher than person B on their well-being (on average), while**
 - ▶ if person A is 20% more talkative today (than they usually are), we predict they will score 20% higher than usual on their well-being (on average).
- ▶ **if person A is 20% more talkative than person B (on average), we predict person A will score 5% higher than person B on their well-being (on average), while**
- ▶ if person A is 20% more talkative today (than they usually are), we predict they will score 10% lower than usual on their well-being (on average).

Example

► Within person effects

- if person A is 20% more talkative than person B (on average), we predict person A will score 5% higher than person B on their well-being (on average), while
- **if person A is 20% more talkative today (than they usually are), we predict they will score 20% higher than usual on their well-being (on average).**

- if person A is 20% more talkative than person B (on average), we predict person A will score 5% higher than person B on their well-being (on average), while
- **if person A is 20% more talkative today (than they usually are), we predict they will score 10% lower than usual on their well-being (on average).**

Model

- ▶ To combine longitudinal and cross-sectional effects, we will include both effects in the model.
- ▶ For example, we can use the linear mixed effects model:

$$Y_{ij} = \mathbf{X}_{ij}^* \boldsymbol{\beta}^{(L)} + \mathbf{X}_{i1}' \boldsymbol{\beta}^{(C)} + \mathbf{W}_i' \boldsymbol{\gamma} + \mathbf{Z}_{ij}' \mathbf{b}_i + e_{ij}$$

where $\mathbf{X}_{ij}^* = \mathbf{X}_{ij} - \mathbf{X}_{i1}$.

- ▶ Here, $\boldsymbol{\beta}^{(C)}$ represent the cross-section effects while $\boldsymbol{\beta}^{(L)}$ are the longitudinal effects.
- ▶ **This is one example, another example is using $\mathbf{X}_{ij}^* = \mathbf{X}_{ij} - \bar{\mathbf{X}}_i$ and $\bar{\mathbf{X}}_i$.**

Interpretation

- For the baseline measurement, it's straightforward to show that

$$E(Y_{i1}) = \mathbf{X}'_{i1}\boldsymbol{\beta}^{(C)} + \mathbf{W}'_i\boldsymbol{\gamma}$$

so $\boldsymbol{\beta}^{(C)}$ is the expected difference in the baseline outcome for a 1 unit change in $\mathbf{X}_{i1}$.

- Further, the model for the within-subject changes is

$$\begin{aligned} E(Y_{ij} - Y_{i1}) &= \mathbf{X}^*_{ij}\boldsymbol{\beta}^{(L)} + \mathbf{X}'_{i1}\boldsymbol{\beta}^{(C)} + \mathbf{W}'_i\boldsymbol{\gamma} - \left(\mathbf{X}^*_{i1}\boldsymbol{\beta}^{(L)} + \mathbf{X}'_{i1}\boldsymbol{\beta}^{(C)} + \mathbf{W}'_i\boldsymbol{\gamma}\right) \\ &= (\mathbf{X}_{ij} - \mathbf{X}_{i1})'\boldsymbol{\beta}^{(L)} \end{aligned}$$

so $\boldsymbol{\beta}^{(L)}$ is the expected within-person change from baseline in the outcome for a 1 unit increase in $\mathbf{X}_{ij}$ from baseline.

Example

- ▶ Let A_{ij} be the age of person i at measurement j and Y_{ij} their BMI. One option is to fit the model

$$Y_{ij} = \beta_0 + \beta_1 A_{ij} + b_{0i} + \epsilon_{ij}$$

- ▶ Separating out the cross-sectional and longitudinal effects of age we have

$$Y_{ij} = \beta_0 + \beta_1^{(L)} A_{ij}^* + \beta_1^{(C)} A_{i1} + b_{0i} + \epsilon_{ij}$$

where $A_{ij}^* = A_{ij} - A_{i1}$ is the change in age from baseline.

Example

- ▶ Separating the cross-sectional and longitudinal effects of age, we have

$$Y_{ij} = \beta_0^* + \beta_1^{(L)} A_{ij}^* + \beta_1^{(C)} A_{i1} + b_{0i}^* + \varepsilon_{ij}.$$

- ▶ For subjects with a baseline age that was 1 unit higher, we expect their baseline outcome to be $\beta_1^{(C)}$ units larger.
- ▶ During our study, if an individual ages 1 year from baseline, the expected change in the outcome is $\beta_1^{(L)}$ units from baseline.

Example

- We can also do this with the average, where our model is

$$Y_{ij} = \beta_0 + \beta_1^{(L)} A_{ij}^* + \beta_1^{(C)} \bar{A}_i + b_{0i} + \varepsilon_{ij}$$

with $\bar{A}_i = \frac{A_{i1} + A_{i2} + \dots + A_{in_i}}{n_i}$ representing a subject's average age over the observed measurements.

- For subjects with an average age that was 1 unit higher, we expect their average outcome to be $\beta_1^{(C)}$ units larger.
- During our study, if an individual ages 1 year, the expected change in the outcome is $\beta_1^{(L)}$ units from their average outcome.

- └ Longitudinal and Cross-sectional effects
 - └ Testing Longitudinal vs Cross-Sectional

$\beta^{(L)}$ versus $\beta^{(C)}$

- When $\beta^{(L)} = \beta^{(C)} = \beta$

$$Y_{ij} = \beta_0^* + \beta_1^{(L)} A_{ij}^* + \beta_1^{(C)} A_{i1} + b_{0i}^* + \varepsilon_{ij}, \quad =$$

$$\beta_0 + \beta_1 A_{ij} + b_{0i} + \varepsilon_{ij}$$

which is the “standard” model with just one Age term.

- So the hypothesis test $H_0 : \beta^{(L)} = \beta^{(C)}$ tests whether longitudinal and cross section effects are equal.

Longitudinal and cross-sectional information

- ▶ Assessment of within-subject changes in the response due to aging (for example) can only be achieved within a longitudinal study design.
- ▶ What would happen with a cross-sectional design?
- ▶ Recall the Muscatine Coronary Risk Factor (MCRF) study, which had five cohorts of children, initially aged 5–7, 7–9, 9–11, 11–13, and 13–15.
- ▶ Goal: determine whether the risk for obesity increases with age.
- ▶ Measurements were taken in 1977, 1979, and 1981.
- ▶ Suppose we have BMI as an outcome of interest.

- └ Longitudinal and Cross-sectional effects
- └ Testing Longitudinal vs Cross-Sectional

Longitudinal and cross-sectional information

- ▶ Could we measure the effect of age using only data from 1977?
- ▶ Differences in $\beta^{(L)}$ versus $\beta^{(C)}$ can arise when there are cohort or period effects.
- ▶ When we assume $\beta^{(L)} = \beta^{(C)} = \beta$, i.e., use the “standard” linear mixed effects model, we get an estimate for β that is a combination of $\beta^{(L)}$ and $\beta^{(C)}$.

Covariance Pattern Analysis vs. Linear Mixed Models

General model:

$$Y_{ij} = X_{ij}\beta + Z_{ij}b_i + \varepsilon_{ij}$$
$$b_i \sim N(0, D), \quad \varepsilon_i \sim N(0, R_i)$$

Typically:

- ▶ **Covariance Pattern Analysis (CPA)**
 - ▶ Classical linear model with an explicit **covariance matrix** for the errors.
 - ▶ No random-effects terms (no b_i); correlation handled solely through $R_i = \text{Cov}(\varepsilon)$.
- ▶ **Linear Mixed Models (LMM)**
 - ▶ Add **random intercepts/slopes**.
 - ▶ Correlation arises from both random effects. Use $R_i = \sigma^2 I$.
- ▶ When should you favor one over the other?

When Is Covariance Pattern Analysis a Good Choice?

- ▶ **Few, fixed time points** (e.g., pre/post or 3–4 visits).
- ▶ **Balanced design** with identical measurement occasions across subjects.
- ▶ Flexibility to **directly specify** a working covariance structure: compound-symmetry, AR(1), unstructured, etc.
- ▶ **Small sample of clusters/subjects**: random-effects variance hard to estimate.
- ▶ **No continuous fixed effects**: if all variables are dummy coded, you can only use a random intercept.
- ▶ **Primary interest = population mean** trend, not subject-specific prediction.

When Do Linear Mixed Models Shine?

- ▶ **Many or irregular time points**; unbalanced follow-up schedules.
- ▶ Need to **model heterogeneity** in trajectories (random slopes, subject-specific deviations).
- ▶ Desire for **individual-level prediction** (e.g., best linear unbiased predictors).
- ▶ **Clustered data beyond subjects over time**: siblings in families, patients in clinics, etc.
- ▶ Allows **cross-level interactions**: e.g., random “teacher effect” $\times$ previous child-level reading score $\Rightarrow$ reading score.

Random Effects *and* Residual Covariance: When Would You Use Both?

Rarely, both are used when:

- ▶ Random effects capture subject-specific mean structure
 - ▶ e.g., random intercepts and slopes
- ▶ *and* there is additional within-subject serial correlation
 - ▶ e.g., AR(1) residual correlation after accounting for random slopes

Examples:

- ▶ Intensive longitudinal data (many time points)
- ▶ Random intercept + slope model *plus* short-term autocorrelation
- ▶ Clustered data with repeated measures within clusters

Why is Random Effects + Structured R_i uncommon in practice?

Key issue: Both random effects and residual covariance structures generate within-subject correlation.

Marginal covariance of Y_i (for mixed models):

$$\text{Var}(Y_i) = Z_i D Z_i^\top + R_i$$

Marginal covariance of Y_i (for covariance pattern):

$$\text{Var}(Y_i) = R_i$$

If both components are used, the model may not be able to distinguish:

- ▶ Correlation induced by random effects ($Z_i D Z_i^\top$)
- ▶ Correlation induced by structured residuals (R_i)

Identifiability issues

Example:

- ▶ Random intercept $\Rightarrow$ constant correlation
- ▶ Residual CS structure $\Rightarrow$ also constant correlation

Consequences:

- ▶ Non-identifiable or weakly identifiable parameters
- ▶ Large standard errors
- ▶ Convergence problems
- ▶ Boundary estimates (variance ≈ 0)

Practical guidance:

- ▶ When using random effects, start with $R_i = \sigma^2 I$
- ▶ Add structure to R_i only if diagnostics strongly suggest additional residual correlation.

Decision Cheat Sheet

	Choose CPA	Choose LMM
Time points	≤ 4 , fixed	Many / irregular
Design	Balanced	Unbalanced
Main target	Population mean	Subject-specific or Population mean
Random heterogeneity	Not of interest	Key feature
Sample size (clusters)	Small	Moderate/large

Rule of thumb: If you have random trajectories or messy follow-up, lean LMM; if you want a clean mean comparison with few waves, CPA is simpler and sufficient.